Published on *Journal of Intelligent Transportation Systems Technology, Planning, and Operations*;
Volume 28, 2024 - Issue 3; Pages 372-385

A Multi-State Merging Based Analytical Model for an Operation Design Domain of Autonomous Vehicles in Work Zones on Two-lane Highways

Qing Tang, Ph.D. Student

Department of Civil and Environmental Engineering

Pennsylvania State University

212 Sackett Building, University Park, PA 16802-1408, USA

Email: qingtang@psu.edu

Xianbiao Hu, Assistant Professor. (Corresponding Author).

Department of Civil and Environmental Engineering

Pennsylvania State University

212 Sackett Building, University Park, PA 16802-1408, USA

Phone: (814) 863-3088

Email: xbhu@psu.edu

Abstract

As a special application of connected and automated vehicles (CAVs), the Autonomous Truck Mounted Attenuator (ATMA) vehicle system is promoted to reduce fatalities in work zone locations. In this manuscript, we focus on the Operational Design Domain (ODD) problem of two-lane highways, i.e., under what traffic conditions should an ATMA be deployed. Due to the dramatic speed difference between ATMA vehicles and general vehicles, a queue will be formed, leading to a percent-time-spent-following (PTSF) increase during maintenance. General vehicles in the queue will assess a gap on the opposite lane to perform a passing maneuver, which is broken down into multi-stage merging behavior. As such, an analytical model is first made, based on queuing theory in which the arrival rate and service rate are analyzed to estimate the PTSF. In this way, the linkage between annual average daily traffic (AADT) and level of service (LOS) is analytically established. Then, the proposed model is validated by comparing the estimated PTSF with that of the Highway Capacity Manual (HCM) values. The comparison results show that the mean error is 9.58%, and the mean absolute error is 12.36%, which demonstrate that the developed model is able to generate satisfactory results when compared with the HCM model. Numeric analysis also shows that roadway performance is sensitive to the K factor and D factor, as well as the operating speed of an ATMA. If LOS=C is a desirable design objective, a good AADT threshold to use would be around 11,000 vehicles per day.

Keywords: Connected and Automated Vehicle (CAV); Autonomous Truck Mounted Attenuator (ATMA); Work Zone; Operational Design Domain (ODD); Level of Service (LOS);

1. INTRODUCTION

Transportation infrastructure maintenance is essential to improve the efficiency of the highway system. The U.S. Department of Transportation (USDOT) spent a total of $24 billion dollars to preserve national highway systems in 2020 alone. In particular, the mobile and slow-moving operations, such as striping, sweeping, bridge flushing, and pothole patching, are critical for the efficient and safe operation of a highway transportation system. Performing the maintenance required for a roadway infrastructure, however, could involve risks, and many work zone crashes have involved State DOT workers. In the State of Missouri, for example, slow-moving operation vehicles have been involved in crashes more than 80 times since 2013, resulting in many injuries to DOT workers (MoDOT 2018). One significant reason for those crashes is the dramatic difference in speed between slow-moving operation vehicles and faster general vehicles. Apart from this reason, aggressive or distracted driving (Harb et al. 2008), the performance of heavy vehicles, speed, and dynamic traffic conditions (Meng and Weng 2011, Ma et al. 2019) may also play a role. How to reduce hazards and achieve a safer environment for DOT workers at work zone locations remains an urgent problem.

As an application of connected and automated vehicles (CAVs), the Autonomous Truck Mounted Attenuator (ATMA) vehicle system is promoted by the Federal Highway Administration (FHWA) and a dozen State DOTs to reduce fatalities at work zone location. The ATMA system in leader-follower style is adopted to remove DOT workers from a following truck during the maintenance of transportation infrastructures, for example in Colorado and Missouri (Descant 2017, Tang et al. 2021). A brief introduction of the ATMA system is presented in Section 2, and can also be found in (Tang et al. 2021, Tang et al. 2021). Meanwhile, one of the major challenges is that the practicable and implementable guidance for ATMA's deployment is practically absent from the *Manual on Uniform Traffic Control Devices* (MUTCD) and other federal regulations and national standards, though the ATMA technology is being rapidly developed and deployed. Thus, state DOTs have tried to develop their own criteria to determine when and where to deploy ATMA. For example, Colorado DOT is using an annual average

65 daily traffic (AADT) of less than 6,000 as the criteria to identify low-volume roads for ATMA
66 deployment. This is due to the nature of mobile and slow-moving operations, ATMA vehicles are usually
67 driving slowly (such as 5~15mph) and, as such, the argument is to avoid slowing traffic on a busy
68 corridor during peak hours. An obvious challenge to this criterion is that AADT on a two-lane highway
69 varies significantly for different DOTs. Roads with an AADT, lower than 6,000, might be common in
70 Colorado while, in other states, such as California and New York, most roads are much busier than that.
71 Thus, the problems are whether this AADT threshold is reasonable, and how should we develop a sound
72 theory to analytically determine this threshold?

73 To bridge this important gap, this manuscript aims to develop a multi-stage querying-based model to
74 analytically identify the traffic Operational Design Domain (ODD) of ATMA on a typical two-lane
75 highway. The USDOT defines ODD as an automated function or system designed to properly operate
76 automated vehicles, including consideration of, but not limited to roadway types, speed range,
77 environmental conditions (weather, daytime/nighttime, etc.), and other domain constraints (NHTSA
78 2016). In this manuscript, we use AADT to reflect the traffic ODD of the ATMA system. The percent-
79 time-spent-following (PTSF), suggested by the *Highway Capacity Manual* (HCM) (FHWA 2010), is
80 chosen as an indicator to measure the level of service (LOS) of a two-lane highway. The PTSF reflects
81 the impact of ATMA vehicles on traffic flow and can be applied to indirectly identify the traffic ODD of
82 ATMA system deployment.

83 In reality, it is difficult to measure PTSF directly because of the complexity of collecting data for faster
84 traveling vehicles that are following slower vehicles. The HCM suggests that PTSF can be estimated by
85 measuring the percentage of vehicles traveling with headways of less than 3s at a representative location
86 on a one-lane highway segment. However, some studies (Luttinen 2001, Dixon et al. 2002) have disputed
87 that the PTSF is overestimated by the HCM. Consequently, many other attempts have been made to
88 estimate the PTSF by analytical models based on different parameters of the traffic stream. Pursula
89 estimated the PTSF based on shockwave analysis (Pursula 1995). He assumed that a traffic stream in an

uncongested regime has two different states--within and outside of platoons--with different densities, which are shockwaves propagating throughout traffic flow with speed. The PTSF was defined by summation of the travel time for all vehicles that travel in platoons, divided by the total travel time within and outside of platoons. Laval applied the kinematic wave theory of moving bottlenecks (Newell 1998) to study the influence that a slow vehicle has on a two-lane highway and derived a formulation to estimate the PTSF (Laval 2006). Three different states for a traffic stream were defined: State A, the traffic flow is not congested; State U, a platoon led by a slow vehicle traveling at a lower speed; and State D, fast vehicles have left the platoon and travel at a free flow speed after passing the slow vehicle. The PTSF was defined as travel time in platoon divided by the total travel time on the segment, which can be derived by travel speed, backward wave speed, and density. Cohen and Polus estimated the PTSF as a function of data, observed in a field scenario, based on queuing theory (Polus and Cohen 2009, Cohen and Polus 2011). Each platoon, caused by a slower vehicle, was considered as a queuing system with a single server, where the service time is the time interval spent by the fast vehicle in the first position, just behind the slower vehicle. The queue was suggested to be described as a M/M/1 queuing model, since the distribution of the number of headways within platoons (collected on highways) was fitted by a geometric distribution. The PTSF was estimated by the average number of headways observed within the platoons and outside the platoons.

The PTSF of a two-lane highway with ATMA vehicles has not yet been explored in any studies. Due to the special leader-follower style, ATMA vehicles are different from general slow-moving vehicles. The speed of ATMA vehicles is usually set as 5~15 mph (Tang et al. 2021), which is obviously much slower than other general vehicles, or even slow-moving trucks. Thus, vehicles impeded by AMTA vehicles are more likely to perform a passing maneuver using the opposite lane. However, in existing research, the probability of performing a passing maneuver is not yet analytically modeled. Besides, the ATMA system contains two vehicles and a certain gap distance between them makes a passing maneuver more complex. On the other hand, before the ATMA system is deployed in a work zone, field data, i.e., travel time, and

115 the number of vehicles within the platoons and outside the platoons, could not be collected to estimate the
116 PTSF using existing methods, and further evaluate the LOS on a two-lane highway.

117 The contributions of this paper include: 1) developing an analytical model to estimate the PTSF of a two-
118 lane highway segment when ATMA vehicles are performing maintenance; 2) identifying the traffic ODD,
119 indicating by AADT threshold in this paper, for ATMA system deployment. The major challenge is to
120 derive the PTSF under the impact of ATMA vehicles on the traffic. Due to significant differences in
121 speed, general vehicles impeded by ATMA vehicles form a queue and are willing to overtake them using
122 the opposite lane. A vehicle in the queue will assess a gap in the opposite lane and, if the gap is
123 acceptable, a passing maneuver will be performed. This process is called "service" in the queue. If we
124 assume that only the first vehicle in the queue, behind ATMA vehicles, has an opportunity to perform a
125 passing maneuver, then this queue can be considered as a queue with one server. Vehicles, or called
126 "customers", join in the queue and travel in the platoon until they move to the first position and find an
127 acceptable gap for performing a passing maneuver. Referring to Cohen and Polus' model (Cohen and
128 Polus 2011), we apply the M/M/1 queueing model to describe the passing process. To derive the service
129 rate, the probability of performing a passing maneuver is analyzed by breaking down a passing process
130 into a multi-stage merging behavior.

131 The arrival rate can also be derived if the flow rate is known. Then, the average wait time in the queue
132 system can be obtained. In addition, with the average travel time between platoons, the PTSF on a one-
133 lane highway segment can be derived. Under different traffic demands on the main lane and the opposite
134 lane, the opportunity for executing a passing maneuver and time headway will change. Thus, the arrival
135 rate and service rate for the queue system will also vary. The PTSF will vary, reflecting the change of
136 LOS with different traffic demand.

137 This manuscript is organized as follows. Some preliminaries are presented in Section 2, including ATMA
138 system overview, performance measurement of two-lane highways, and the M/M/1 queuing model.
139 Section 3 presents the model development, in which the PTSF of a two-lane highway segment, with

ATMA vehicles and without ATMA vehicles, are analytically derived. The proposed model is validated by comparing the results with that of HCM model, and the modeling result analysis and sensitivity analysis are discussed in Section 4. Section 5 concludes this manuscript.

2. PRELIMINARIES

In this section, the ATMA system is briefly introduced at first, followed by a review of performance measurements of two-lane highways from HCM. Then, we review the M/M/1 queuing model to describe a queue with a single server which serves customers in the order in which they arrive. The acronyms and notations used in this manuscript are summarized in Table 1 and

Table ***2***.

Table 1 List of acronyms used in the paper

AADT	Average Annual Daily Traffic
ATMA	Automated Truck Mounted Attenuator
ATS	Average Travel Speed
CAV	Connected and Automated Vehicle
DOT	Department of Transportation
ODD	Operational Design Domain
FHWA	Federal Highway Administration
FT	Follower Truck
HCM	Highway Capacity Manual
LOS	Level of Service
LT	Leader Truck
MUTCD	Manual on Uniform Traffic Control Devices
PTSF	Percent Time Spent Following

Table 2. List of notations used in the paper

t	the time between arrivals in a queue

λ	arrival rate
$f(t)$	probability density function of the time between arrivals
s	service time of a queue
μ	service rate
$g(s)$	probability density function of service time
ρ	traffic intensity
Q	average number of customers in the system
W	average time spent in the system
L	length of the analyzed two-lane highway segment
L_{gap}	gap distance between the leader truck and the follower truck
L_{lt}	length of leader truck or follower truck
v_{lt}	travel speed of ATMA vehicles, i.e., leader truck and follower truck
v_{u}	cruising speed of general vehicles
v_{gt}	travel speed of general trucks
a	acceleration of general vehicles
$\bar{h}$	average time headway on the analyzed lane
Δt	inter-arrival times between general vehicles at the back end of the platoon
L_2	the distance between the first vehicle in the queue and the leader truck when the vehicle returns to the original lane
L_3	the distance that the first vehicle in the queue traveled on the opposite lane
t_{pass}	the average time for passing
t_{AB}, t_{BC}, t_{CD}	travel time from Point A to B, Point B to C, and Point C to D, respectively
h'	time headway on the opposite lane
λ'	the number of vehicles arriving in a specific time interval on the opposite lane

$f(h', \lambda')$,	probability density function of the time headway
$F(h', \lambda')$	cumulative distribution of the time headway
P_{pass}	the probability of performing a passing maneuver for the first vehicle in the queue
$\bar{h}'$	average time headway
$\bar{t}_s$	average service time for the first vehicle in the queue
T	average travel time of general vehicles between platoons with the cruising speed
N_{gt}	number of general trucks

2.1 ATMA Introduction

The ATMA system is a quickly emerging technology and offers a promising solution for eliminating injuries to DOT workers while performing roadway maintenance. Figure 1 shows pictures of an ATMA vehicle system under operation on a roadway, which include (a) an overview of the system, (b) a view from above, and (c) a view from the rear. The vehicle system consists of a manned leader truck, an unmanned follower truck, and a truck mounted attenuator (TMA) installed on the follower truck. The leader truck is designed to perform the regular maintenance work, whereas the follower truck is designed to drive autonomously and duplicate the behavior of the leader truck, as well as to serve as a buffer in case a crash occurs. The gap distance between the leader truck and the follower truck is usually pretty small (e.g., 100~200 ft). As a result, the leader-follower design significantly simplifies the self-driving scenario and allows the follower truck to simply mimic the action of the leader truck. The idea is that, in case, a highway crash is inevitable at a work zone location, the follower truck will be hit first and, since it drives autonomously without a human driver and is equipped with TMA, the DOT workers' lives will be protected. Actuators, software, electronics, and vehicle-to-vehicle (V2V) communication equipment are installed on the leader truck and the follower truck and, together, they enable connectivity (mainly V2V) and the autonomous driving capabilities of a leader-follower style.



(a)



(b)



(c)

Figure 1 ATMA Vehicle System: (a) system overview; (b) view from above; and (c) rear view

2.2 Performance measurements of two-lane highways

The LOS is a quantitative stratification of performance measures that represents the quality of service. Three measures of effectiveness are used to determine an automobile LOS on two-lane highways from HCM (HCM 2010), i.e., average travel speed (ATS), percent time-spent-following (PTSF), and percent of free-flow speed (PFFS). The HCM defines three classes of two-lane highways, in which Class I is a highway where motorists expect to travel at relatively high speeds, Class II is a highway where motorists do not necessarily expect to travel at high speeds, and Class III is a highway serving moderately developed areas. In this manuscript, we mainly focus on the Class I two-lane highway scenario, in which PTSF and ATS are two measures to determine the LOS. PTSF is the average percentage of time that vehicles must travel in platoons behind slower vehicles due to the inability to pass. As PTSF is difficult to measure, a surrogate measure, suggested by the HCM, is the percentage of vehicles traveling at headways of less than 3s at a representative location within a highway segment. ATS is defined as a highway

segment length, divided by the average travel time taken by vehicles to traverse it, during a designated time interval.

Figure 2 describes how ATS and PTSF determine the LOS for two-lane highways from the HCM. For Class I two-lane highways, this figure can be used to determine the LOS of any traffic flow, once the ATS and PTSF are given. Although the HCM and other studies have proposed several methods for estimating the PTSF, the traffic environment is different after involving ATMA vehicles due to the dramatic differences in speed. Moreover, filed data for ATMA vehicles is absent so far, and the ATS and PTSF cannot be obtained directly. As discussed before, when platoons generated by ATMA vehicles or slower general trucks are formed, the result is an obvious increase in the PTSF of general vehicles. Those general vehicles in the queue will try to find an acceptable gap on the opposite lane to execute a passing maneuver. A queuing model is proposed in Section 3, to analytically derive the mean arrival time and mean service time of the queue, generated by ATMA vehicles or general trucks, and then the average waiting time in the queue can be derived. Thus, the PTSF, affected by ATMA vehicles and general trucks can be estimated, and can be compared with the thresholds in Figure 2 to determine the LOS of the two-lane highway segment.

LOS	Class I Highways		Class II Highways	Class III Highways
	ATS (mi/h)	PTSF (%)	PTSF (%)	PFFS (%)
A	>55	≤35	≤40	>91.7
B	>50–55	>35–50	>40–55	>83.3–91.7
C	>45–50	>50–65	>55–70	>75.0–83.3
D	>40–45	>65–80	>70–85	>66.7–75.0
E	≤40	>80	>85	≤66.7

Figure 2 The LOS for Two-lane Highways from HCM (2010).

2.3 M/M/1 queuing model

The flow on a two-lane highway is different from the flow on multi-lane highways, mainly because vehicles need to consider the traffic flow in the opposite lane when they are impeded by slow-moving vehicles. Vehicles may be subject to delays because of uncertain opportunities to pass slow-moving

vehicles by using the opposite lane. Those vehicles, impeded by slow-moving vehicles, will form platoons, and the length of a platoon depends on the traffic demand from each direction, while other vehicles can travel between platoons at cruising speed. Vehicles, traveling in platoons, evaluate gaps on the opposite lane and assess whether they can successfully complete a pass, or not, so they make a decision as to whether they should overtake a leading slow-moving vehicle. Vehicles behind the slow-moving vehicle form a moving queue and, thus, these behaviors can be analyzed by queuing theory models with some necessary modifications. The second vehicle in the platoon behind the slow-moving vehicle is also the first vehicle in the queue, and the service is the time that this vehicle spends in the first position just behind the slow-moving vehicle.

One queuing models is adopted to describe the above traffic flow behavior in this manuscript. The M/M/1 queuing model describes a queue with a single server which serves customers in the order in which they arrive. The number of customers is assumed to follow the Poisson distribution, which means the time between arrivals, t, has an exponential distribution with parameter λ, i.e., the probability density function is $f(t) = \lambda \mathrm{e}^{-\lambda t}$. The service time, s, is also assumed to follow exponential distribution with parameter μ, i.e., the probability density function is $g(s) = \mu e^{-\mu s}$. The service times for different customers are independent, and customers are served in their order of arrival, i.e., first-in-first-out (FIFO) service discipline. The traffic intensity, defined as $\rho = \lambda/\mu$, is a measure of how busy a system is, and is assumed to be less than 1 to make sure the queue is stable. The average number of customers in the system is given by Eq. (1).

$$Q = \Sigma_{n=0}^{\infty} n P_n = \frac{\lambda}{\mu - \lambda} = \frac{\rho}{1 - \rho} \tag{1}$$

The time spent in the system W can be described as negative exponential distribution $\mu - \lambda$. The expected time spent in the system is derived as

$$E(W) = \int_0^{\infty} w(\mu - \lambda) e^{-(\mu-\lambda)w} dw = \frac{1}{\mu - \lambda} \quad (2)$$

3. ANALYTICAL DERIVATION OF PERFORMANCE MEASURES

In this section, we analytically derive the PTSF on a typical two-lane highway scenario. We start with describing the problem as a one-server queueing system. The arrival rate and the service rate are then analytically derived to obtain the average time spent in the system. At last, the PTSF of the typical two-lane highway is estimated from the PTSF affected by ATMA vehicles and general slow-moving trucks.

3.1 Problem setup

First, we focus on a typical two-lane highway segment with length, L, one lane in each direction, as shown in Figure 3. The ATMA vehicle system is represented by a leader truck (LT) and a follower truck (FT), with a gap distance of L_{gap} between these two vehicles. The leader truck is responsible for performing maintenance and the unmanned follower truck provides a safety guarantee. ATMA vehicles enter the segment at a constant speed v_{lt} and leave the segment after completing maintenance. General vehicles, presented by a smaller black or white vehicle icon, travel at a cruising speed v_u before encountering slow-moving ATMA vehicles. Several general vehicles join a platoon (shown in red box in Figure 3), and slow down to v_{lt} once they encounter ATMA vehicles. The platoon can be considered as a one-server queuing system, in which service time is the time spent by the first vehicle in the queue just behind the ATMA vehicles. General vehicles, that join the back of the platoon, are the customers in this one-server queuing system. The PTSF, affected by ATMA vehicles, is the percentage of time that general vehicles travel in the platoon, behind the ATMA, before they find an acceptable gap in the opposite lane to execute a pass maneuver. In order to estimate the PTSF, we need to clarify how to analyze the arrival rate and the service rate of this queue and then derive the average time spent in the platoon.

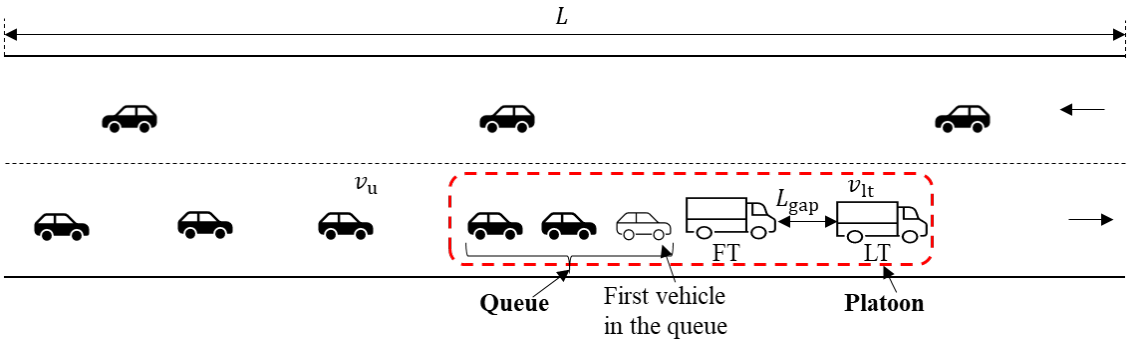


Figure 3 Schematic diagram of a two-lane highway segment with ATMA vehicles.

3.2 Derivation of the PTSF estimates

In this section, we show how to analytically derive the PTSF estimates. First, we adopt some assumptions in a theoretical model proposed by Cohen and Polus (Polus and Cohen 2009, Cohen and Polus 2011), including: (1) The speeds of fast and slow vehicles are assumed to be constant, except during the overtaking process; (2) The flow of traffic is along an infinite homogeneous section, without any entrances or exits and with similar geometry. No-passing zones are neglected in the model; (3) The flow operates in a steady-state condition, which means that the mean service time should not be greater than the mean inter-arrival time because, as such, the queue will end, in the long term, with an infinite queue. In addition, two more assumptions are made in this manuscript:

(1) The drivers of general vehicles are willing to pass ATMA vehicles, using the opposite lane, once conditions are satisfactory.

(2) Since the ATMA system includes two trucks and a gap distance between them, it is more difficult for general vehicles to pass. We assume that only one general vehicle can perform a passing maneuver each time, which means the first vehicle in the queue (shown in Figure 3) will pass the ATMA vehicles first.

In Figure 3, general vehicles trail one another in a platoon because these vehicles are impeded by ATMA vehicles and it is difficult to perform a passing maneuver because of on-coming traffic. The general vehicles that are impeded by the ATMA vehicles form a queue and travel at a speed of v_{lt}. The queue in

Figure 3 is considered as a one-server queuing system, in which service time is the time interval spent by the vehicle in the first position in the queue until this vehicle finds an acceptable gap and performs a passing maneuver. Once a general vehicle becomes the first vehicle in the queue, just behind the follower truck, that vehicle will find an acceptable gap on the opposite lane to perform a passing maneuver. Accordingly, we adopt the M/M/1 queueing model, and analytically derive the arrival rate and service rate. The PTSF is then estimated by the average waiting time of general vehicles in the platoon.

3.2.1 Derivation of arrival rate λ

First, we derive the arrival rate for the platoon generated by the slow-moving ATMA vehicles. The customers in this one-server queueing system are the general vehicles joining the back of the platoon. Figure 4 presents a situation where the inter-arrival time of the general vehicles at the back of the queue is schematically shown. At time t, a general vehicle (white general vehicle icon) travels at a speed of v_{u}, and the distance between it to the last vehicle in the platoon is $v_{\mathrm{u}} \cdot \bar{h}$, in which $\bar{h}$ is the average time headway on the analyzed lane. At time $t + \Delta t$, this general vehicle joins the platoon. As such, we have $v_{\mathrm{u}} \cdot \Delta t = v_{\mathrm{u}} \cdot \bar{h} + v_{\mathrm{lt}} \cdot \Delta t$. The inter-arrival times between general vehicles at the back end of the platoon is given by Eq.(3).

$$\Delta t = \frac{v_{\mathrm{u}} \cdot \bar{h}}{v_{\mathrm{u}} - v_{\mathrm{lt}}} \tag{3}$$

Thus, the arrival rate of general vehicles can be derived by Eq.(4).

$$\lambda = \frac{1}{\Delta t} = \frac{v_{\mathrm{u}} - v_{\mathrm{lt}}}{v_{\mathrm{u}} \cdot \bar{h}} \tag{4}$$

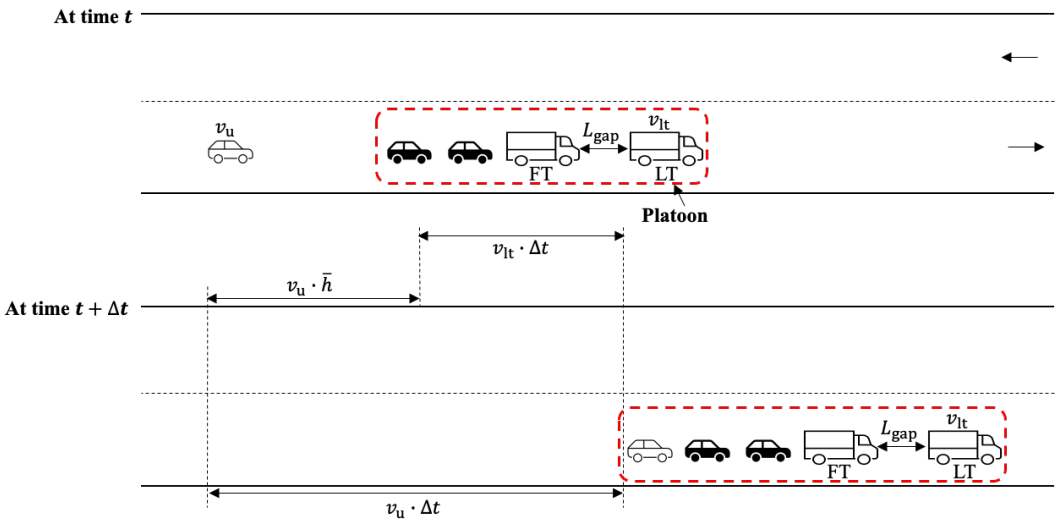


Figure 4 Two-time snapshots of the ATMA platoon and a joining general vehicle.

3.2.2 Derivation of service rate μ with multi-stage merging behavior

Next, we derive the service rate for the platoon generated by the slow-moving ATMA vehicles. Based on the assumption we made, once a general vehicle becomes the first vehicle in the queue, just behind the follower truck, the vehicle will find an acceptable gap on the opposite lane to perform a passing maneuver. As such, the service time of this one server queueing system is the time interval spent by a general vehicle (depicted as a red icon in Figure 5), from this vehicle arriving at the first position in the queue, to start a passing maneuver. In this section, we first analyze the time needed for the first vehicle in the queue to pass the ATMA vehicles, if an acceptable gap is found. The probability of passing and service time are then derived under a certain distribution of time headway on the opposite lane. The passing process of the first vehicle in the queue, shown in Figure 5, is broken down into three stages:

Stage 1: The first vehicle in the queue, behind the follower truck, finds an acceptable gap in the opposite lane and changes lane, from Point A to Point B, at a speed of v_{lt}. The travel length is L_1 and the travel time is denoted by t_{AB}.

Stage 2: The vehicle accelerates to a cruising speed v_{u} with acceleration a, and then travels on the opposite lane until it overtakes the leader truck at a certain distance (i.e., L_2 shown in Figure 5), from

297 Point B to Point C. The total travel length on the opposite lane is L_3 and total travel time is denoted by

298 t_{BC}.

299 *Stage 3*: The vehicle returns to the original lane at a speed of v_{u} from Point C to Point D. The travel

300 length is L_1 and travel time is t_{CD}.

301 Thus, the total time for the first vehicle in the queue to complete a passing maneuver can be described as

302 time spent above the three stages shown in Eq. (5).

$$t_{\mathrm{pass}} = t_{AB} + t_{BC} + t_{CD} \tag{5}$$

303 For the first stage, AB, and third stage ,CD, the time can be calculated by Eqs. (6) and (7).

$$t_{AB} = \frac{L_1}{v_{\mathrm{lt}}} \tag{6}$$

$$t_{CD} = \frac{L_1}{v_{\mathrm{u}}} \tag{7}$$

304 where $L_1$is the lane width.

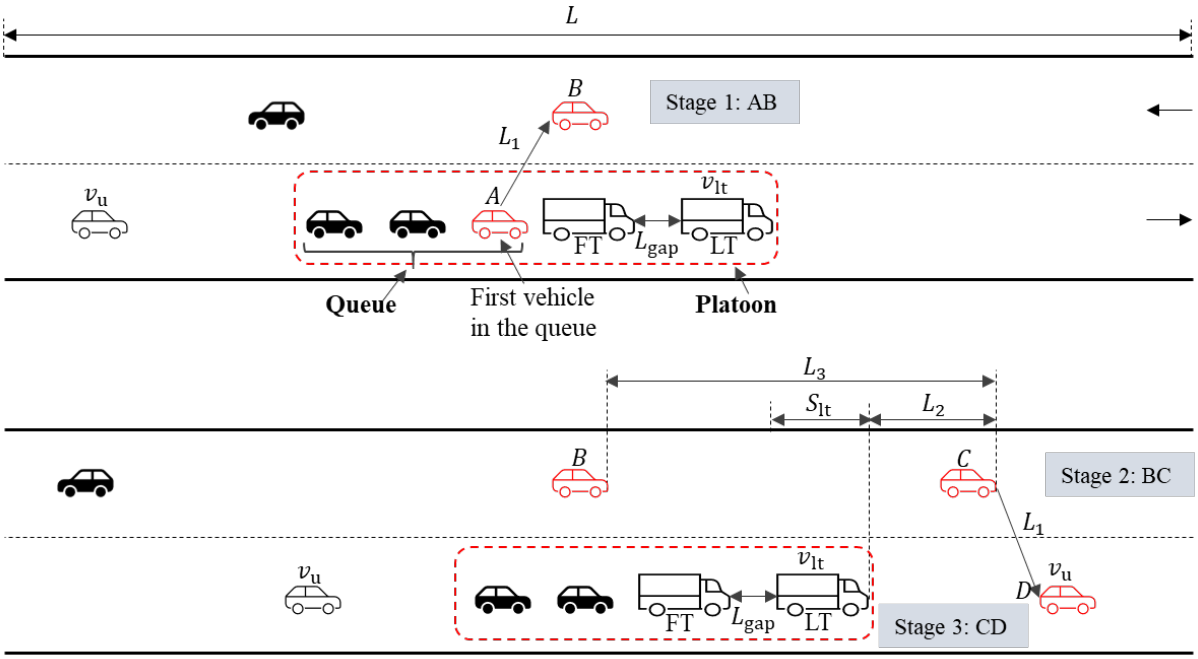


Figure 5 Schematic diagram of the first vehicle in a queue passing ATMA vehicles.

For the second stage, BC, the distance that the vehicle travels is denoted by L_3. The vehicle first accelerates from v_{lt} to a cruising speed v_u, and then travels to Point C at the cruising speed. So, the distance L_3 can be derived from Eq. (8).

$$L_3 = \frac{v_u^2 - v_{lt}^2}{2a} + v_u\left(t_{BC} - \frac{v_u - v_{lt}}{a}\right) \tag{8}$$

On the other hand, the distance ATMA vehicles traveled is $S_{lt} = v_{lt}t_{BC}$ during the second stage BC. The vehicle travels on the opposite lane until it overtakes the leader truck for a certain distance L_2. Therefore, the distance L_3 can also be calculated by Eq.(9).

$$L_3 = v_{lt}t_{BC} + 2L_{lt} + L_{gap} + L_2 \tag{9}$$

where L_{lt} is the length of leader truck or follower truck, L_{gap} is the gap distance between the leader truck and follower truck, and L_2 is the safety distance to make a lane changing. According to Eqs. (8) and (9), the time t_{BC} can be derived by Eq. (10).

$$t_{\mathrm{BC}} = \frac{2L_{\mathrm{lt}} + L_{\mathrm{gap}} + L_2}{v_{\mathrm{u}} - v_{\mathrm{lt}}} + \frac{v_{\mathrm{u}} - v_{\mathrm{lt}}}{2a} \tag{10}$$

So far, we know the total time needed to pass the ATMA vehicles, once the on-coming traffic is satisfied. Next, we analyze the probability of the first vehicle in the queue performing a passing maneuver behind the follower truck. When the traffic conditions were light, or moderate, the time headway h' on the opposite lane can be described by negative exponential distribution, as shown in Eqs. (11) and (12).

$$f(h', \lambda') = \begin{cases} h' e^{-\lambda' t} & h' \geq 0, \\ 0 & h' \leq 0. \end{cases} \tag{11}$$

$$F(h', \lambda') = \begin{cases} 1 - e^{-\lambda' h'} & h' \geq 0, \\ 0 & h' \leq 0. \end{cases} \tag{12}$$

where $f(t, \lambda')$ is the probability density function, and $F(t, \lambda')$ is the cumulative distribution function. λ' is the number of vehicles arriving at a specific time interval on the opposite lane. The probability for the first vehicle in the queue performing a passing maneuver is denoted by P_{pass}, and can be derived by Eq. (13).

$$P_{\mathrm{pass}} = P\left(h' \geq t_{\mathrm{pass}}\right) = e^{-\lambda' t_{\mathrm{pass}}} \tag{13}$$

Once a general vehicle reaches the first position in the queue, the service starts until the vehicle finds an acceptable gap and leaves the queue to pass the ATMA vehicles. The average service time $\bar{t}_s$ for the first vehicle in the queue can be derived by Eq. (14).

$$\bar{t}_s = P_{\text{pass}} \times 0 + \left(1 - P_{\text{pass}}\right) \times \left\{\bar{h}' + P_{\text{pass}} \times 0 + \left(1 - P_{\text{pass}}\right) \times \left[\bar{h}' + P_{\text{pass}} \times 0 + \cdots\right]\right\}$$
$$= \frac{1 - P_{\text{pass}}}{P_{\text{pass}}} \cdot \bar{h}' \tag{14}$$

where $\bar{h}'$ is the average time headway on the opposite lane. Then, service rate μ can be calculated by Eq. (15).

$$\mu = \frac{1}{\bar{t}_s} = \frac{P_{\text{pass}}}{(1 - P_{\text{pass}}) \cdot \bar{h}'} \tag{15}$$

3.2.3 PTSF estimates

In this section, we derive the PTSF affected by slow-moving ATMA vehicles and general trucks. HCM (HCM 2010) defined the PTSF as the average percentage of time that vehicles must travel in platoons behind slower vehicles due to the inability to pass. Cohen and Polus (Cohen and Polus 2011) defined $\text{PTSF} = \frac{W}{W+T} \times 100\%$, where T is average travel time of general vehicles between platoons. Similarly, the PTSF affected by an ATMA platoon is defined as Eq.(16).

$$\text{PTSF}_{\text{lt}} = \frac{E(W)}{E(W) + T} \times 100\% \tag{16}$$

where T is the average travel time of general vehicles between platoons with a cruising speed v_u, and $E(W)$ is the average time that general vehicles spend in a queueing system. According to Eq.(2), the average time in system $E(W)$, i.e., average time of traveling in the queue from the moment of joining the back of the platoon until the moment of starting the passing maneuver, is given by Eq. (17).

$$E(W) = \frac{1}{\mu - \lambda} = \frac{1}{\dfrac{P_{\text{pass}}}{(1 - P_{\text{pass}}) \cdot \bar{h}'} - \dfrac{v_{\text{u}} - v_{\text{lt}}}{v_{\text{u}} \cdot \bar{h}}} \tag{17}$$

With the assumption of an equilibrium state, the number of general trucks is a constant denoted by N_{gt}. The number of platoons led by ATMA vehicles and general trucks is $N_{\mathrm{gt}}+1$; thus, the average travel time of general vehicles between platoons can be estimated by $T=\frac{L}{N_{\mathrm{gt}}+1}$ for a two-lane highway segment. Thus, the PTSF estimates affected by ATMA vehicles is derived.

Apart from the ATMA platoon, the platoons affected by general slow-moving trucks are also analyzed. Similar with the ATMA platoon, the arrival rate of general vehicles for a general truck platoon can be derived by Eq. (18).

$$\lambda=\frac{1}{\Delta t}=\frac{v_{\mathrm{u}}-v_{\mathrm{gt}}}{v_{\mathrm{u}}\cdot\bar{h}} \tag{18}$$

The service rate of general vehicles for a general truck platoon can also be derived according to Eqs. (5)~(15). The difference is calculating the average time for passing the general truck. For the first stage AB, the time can be calculated by Eq.(19).

$$t_{\mathrm{AB}}=\frac{L_1}{v_{\mathrm{gt}}} \tag{19}$$

According to Eqs. (8) and (9), the time t_{BC} can be derived from Eqs. (20) and (21).

$$L_3=\frac{v_{\mathrm{u}}^2-v_{\mathrm{gt}}^2}{2a}+v_{\mathrm{u}}\left(t_{\mathrm{BC}}-\frac{v_{\mathrm{u}}-v_{\mathrm{gt}}}{a}\right)=v_{\mathrm{gt}}t_{\mathrm{BC}}+L_{\mathrm{gt}}+L_2 \tag{20}$$

$$t_{\mathrm{BC}}=\frac{\mathrm{L_{gt}}+L_2}{v_{\mathrm{u}}-v_{\mathrm{gt}}}+\frac{v_{\mathrm{u}}-v_{\mathrm{gt}}}{2a} \tag{21}$$

Thus, the average time for passing the general truck can be calculated according to Eq.(5). Then, the probability for the first vehicle in the queue performing a passing maneuver, as well as the average

service time and service rate, can be calculated according to Eqs. (13)~(15). The PTSF for a general truck platoon $\mathrm{PTSF}_{\mathrm{gt}}$ can be derived accordingly.

Consequently, the PTSF for mixed platoons, on a typical two-lane highway segment, can be estimated by Eq.(22).

$$\mathrm{PTSF} = \frac{N_{\mathrm{gt}} \cdot \mathrm{PTSF}_{\mathrm{gt}} + \mathrm{PTSF}_{\mathrm{lt}}}{N_{\mathrm{gt}} + 1} \tag{22}$$

4. NUMERICAL ANALYSIS

In this section, we first seek to validate this proposed model by comparing its results with those of the HCM model. Then, we apply the proposed model to a typical ATMA operating condition, with two maintenance truck in lead-follower style, to determine its suitable ODD. Finally, we perform sensitivity analysis of the proposed model.

4.1 Performance measurement validation

To validate our proposed model, we assume a scenario in which a platoon is generated by a slow-moving truck on a two-lane highway segment. In this situation, we can individually apply the method provided by the HCM and our proposed model to estimate the PTSF. As such, the validity of our proposed model is verified through comparison. Figure 6 describes the schematic diagram of the first vehicle (denoted by a red vehicle icon), in the queue affected by a general truck, executing a passing maneuver. The platoon consists of a slower general truck, first vehicle behind the truck, and several general vehicles (denoted by black vehicle icons). The vehicles out of the platoon, that are denoted by white vehicle icons, are traveling at cruising speed v_u. The characteristics for the analyzed two-lane highway segment are as follows: the length of highway segment is $L = 1$ mile without no-passing zones; the lane width is set to be L_1 =12 ft; road condition is level terrain; and no impediments to through traffic (e.g., traffic signals, turning signals). The demand flow rate v for both directions is set as 500~$1{,}500$ veh/hr, and the D-factor is set as 0.55. Then, the directional demand flow rate v_d ranges from 275 veh/hr to 825 veh/hr, and the opposing

flow rate v_0 ranges from 225 veh/hr to 675 veh/hr. The percentage of general trucks is set as 20%. The cruising speed of general vehicles is $v_{\mathrm{u}} = 65$ mph. The cruising speed of trucks is $v_{\mathrm{tr}} = 50$ mph, and the length of general trucks is $L_{\mathrm{truck}} = 48$ ft. The acceleration is set as $a = 14.8\ \mathrm{ft/s^2}$. The safe distance for a vehicle executing a passing maneuver and returning to its original lane is set as $L_2 = 90$ ft, almost five lengths of a general vehicle.

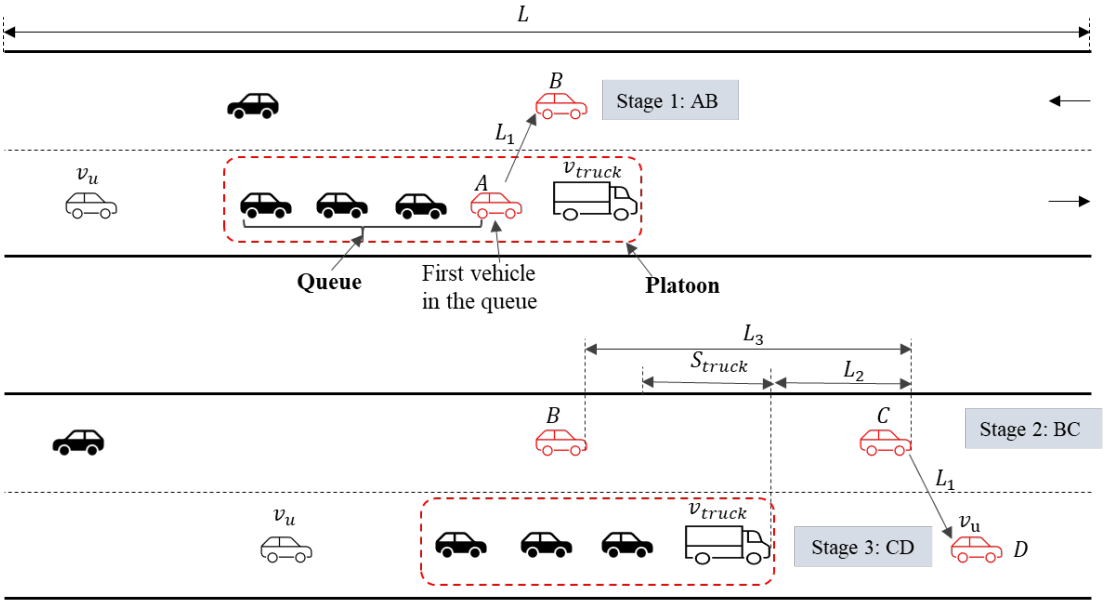


Figure 6 Schematic diagram of the first vehicle in the queue passing a general truck.

The base percent time-spent-following (BPTSF) is estimated by Eq. (23) proposed by HCM (HCM 2010).

$$BPTSF_d = 100[1 - e^{-av_d^b}] \tag{23}$$

where v_d is the directional demand flow rate. The coefficient a and b are constants determined by the opposing demand flow rate v_o and are given in Exhibit 15-20 (HCM 2010).

Table 3 and Figure 7 present the proposed relationship between two-way traffic flow and PTSF estimates, and a comparison with HCM values. The PTSF turns out to increase as the traffic demand is increasing with both methods (shown in Figure 7). The PTSF of our proposed model ranges from 14.5% to 80.6%,

while the PTSF of the HCM model ranges from 29.0% to 69.2%. The overall PTSF difference between HCM values and our proposed model is less than 0 before the traffic flow reaches 1,400 veh/hr. That is to say the PTSF of our proposed model are mostly lower than that of the HCM model. As discussed in Section 1, some studies (Luttinen 2001, Dixon et al. 2002) have disputed that the PTSF is overestimated by the HCM. Thus, it is reasonable that the PTSF by our proposed model is mostly lower than the PTSF of the HCM model. The developed model is able to generate satisfactory results when compared with the HCM model. Once the traffic flow increases beyond 1,400 veh/hr, the passing opportunity will drop significantly, which leads to a dramatic increase in the PTSF.

Table 3 Comparison of PTSF between HCM values and the proposed model

v(veh/hr)	v_d(veh/hr)	v_o(veh/hr)	PTSF (%)		Difference
			HCM	Proposed model	
500	275	225	29.0	14.5	-14.5
600	330	270	34.8	18.3	-16.5
700	385	315	40.2	22.6	-17.6
800	440	360	45.0	27.4	-17.6
900	495	405	49.2	32.7	-16.5
1000	550	450	53.7	38.6	-15.1
1100	605	495	57.5	45.2	-12.3
1200	660	540	60.7	52.6	-8.1
1300	715	585	63.4	60.9	-2.5
1400	770	630	66.3	70.2	3.9
1500	825	675	69.2	80.6	11.4

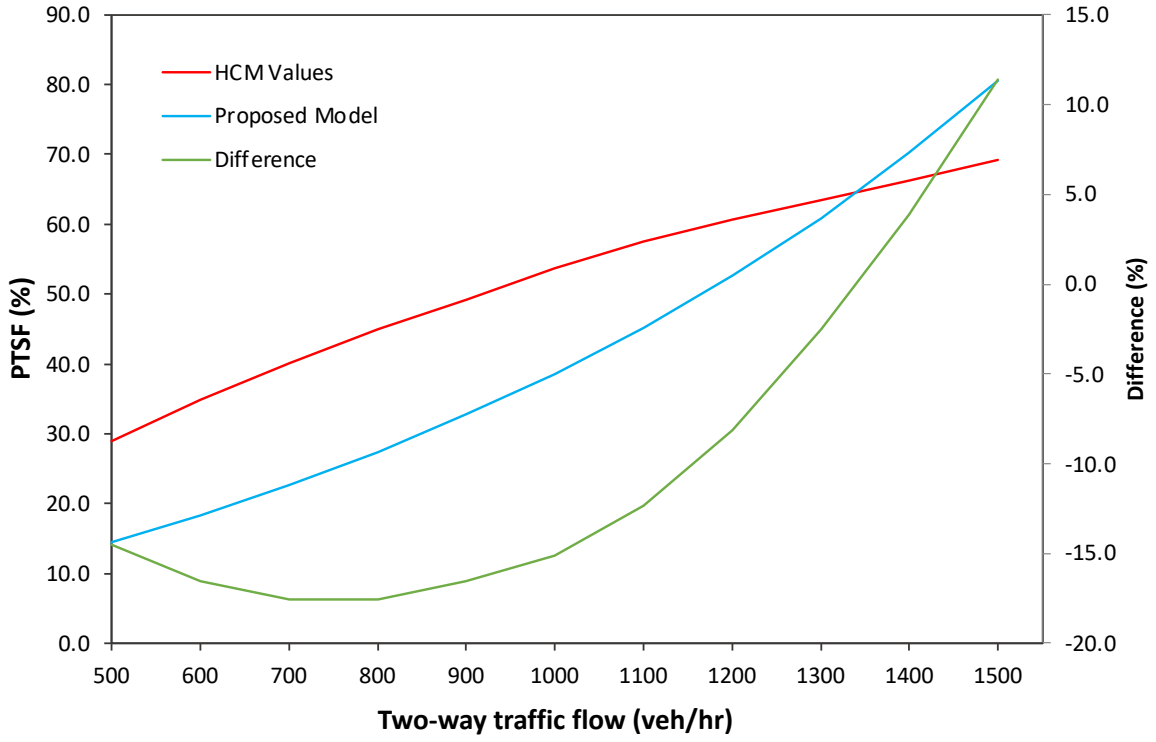


Figure 7 Proposed relationship between two-way traffic flow and PTSF, and a comparison with HCM values.

4.2 Typical scenario ODD analysis

In this section, we analyze the impact of the ATMA system on the PTSF of a typical two-lane highway to identify the potential traffic ODD of this technology. The analysis target is a two-lane highway segment (as illustrated in Figure 5). The general characteristics of this analyzed segment are set as follows: $L = 1$ mile, $v_{\mathrm{u}} = 65$ mph, $v_{\mathrm{truck}} = 50$ mph, $L_{\mathrm{truck}} = 48$ ft, $v_{\mathrm{lt}} = 15$ mph, $L_{\mathrm{lt}} = L_{\mathrm{ft}} = 30$ ft, $L_1 = 12$ ft, $a = 14.8\ \mathrm{ft/s^2}$. For safety, the platoon generated by a general truck, L_2 is set as 90 ft, the length of nearly to five general vehicles. While for a ATMA moving bottleneck, the gap distance between the leader truck and the follower truck is set as $L_{\mathrm{gap}} = 100$ ft, and L_2 is set as 0 because of the dramatic speed difference between ATMA vehicles and general vehicles. Once a vehicle overtakes the ATMA vehicles on the opposite lane, the vehicle will return to its original lane as soon as possible. The percentage of general trucks is assumed to be 20% in this manuscript. Traffic volume on two-lane highways vary widely and

AADT can be up to 36,000 veh/day (HCM 2010). We set the AADT as ranging from 5,000 veh/day to 12,000 veh/day in this manuscript. Recommended by HCM, the values of representative K and D factors are set as 0.09 and 0.55, respectively. The hourly traffic demand then can be estimated by Eq. (24).

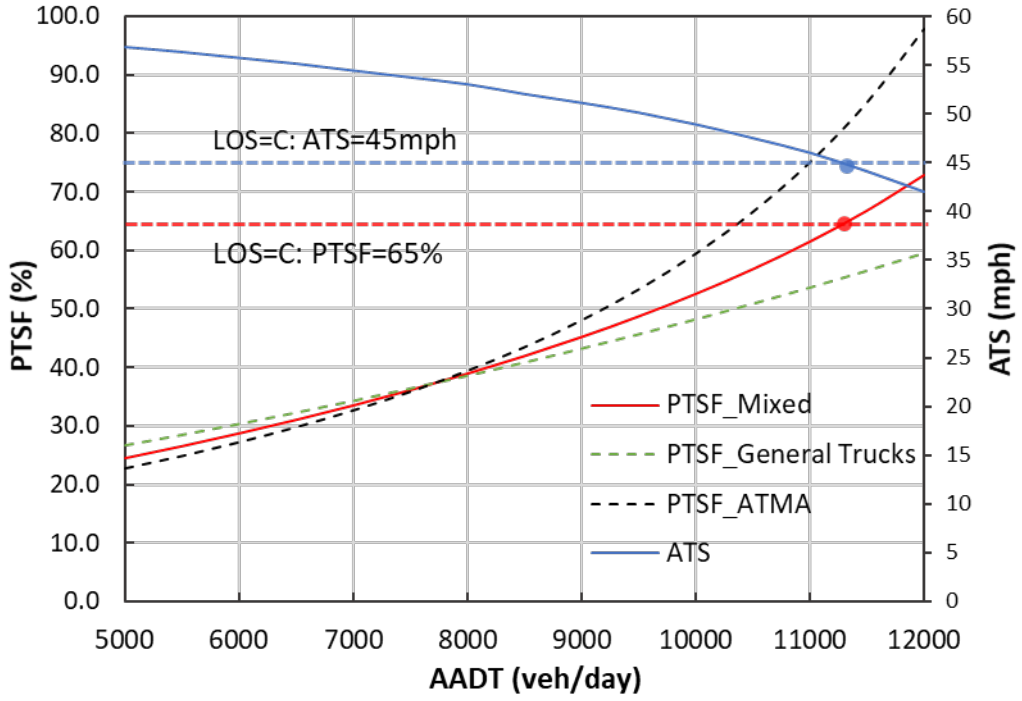
$$v = DDHV = AADT * K * D \tag{24}$$

Based on a setup of highway geometry and traffic flow characteristics, the proposed relationship between traffic demand (i.e., AADT) and PTSF is shown in Figure 8. The PTSF affected by general trucks (denoted by green dotted line in Figure 8), increases from 26.8% to 59.4% as AADT increments from 5,000 veh/day to 12,000 veh/day, while the PTSF affected by ATMA vehicles (denoted by black dotted line) increases sharply from 22.8% to 97.7%. Before the AADT exceeds 7,500 veh/day, the PTSF affected by general trucks is slightly higher than that caused by ATMA vehicles. The times for passing ATMA vehicles and for passing a general truck are constant, based on our proposed model and predefined characteristics. Moreover, the time for passing ATMA vehicles is less than that for passing a general truck, due to the ATMA vehicles' slower speed, which leads to a service rate in an ATMA platoon that is greater than that in a general truck platoon. On the other hand, the arrival rate in a platoon increases as the traffic demand increases, and that for an ATMA platoon increases faster than that for the general truck platoon, due to the ATMA vehicles' slower speed, according to Eq. (4). Thus, the difference between the service rate and the arrival rate for the ATMA platoon is greater than that for the general truck platoon before the AADT reaches 7,500 veh/day. After the AADT exceeds 7,500 veh/day, the difference for the ATMA platoon will less than that for the general truck platoon due to the sharply increasing arrival rate for the ATMA platoon. Therefore, the average waiting time in the ATMA platoon is slightly higher than that in the general truck platoon before the AADT exceeds 7,500 veh/day while, after that, the average waiting time in the ATMA platoon increases dramatically.

The PTSF for mixed traffic (denoted by the red line in Figure 8) increases from 24.5% to 73.0%. The corresponding ATS (described in the blue line) decreases from 57 mph to 42 mph. The PTSF thresholds

435 that differentiate LOS are 35%, 50%, 65%, and 80%, respectively, while the ATS thresholds that
436 differentiate LOS are 55 mph, 50 mph, 45 mph, and 40 mph. We use the PTSF and ATS to define the
437 LOS category in Figure 8 as well. The desirable design objective is set as LOS=C for ATMA's operation
438 design domain, which corresponds to between 11,000 and 11,500 AADT in Figure 8. As such, for this
439 typical scenario, the AADT of 11,000 veh/day is identified as a reasonable threshold for defining
440 ATMA's ODD.

441



442

443 **Figure 8 Proposed relationship between AADT and PTSF (and ATS).**

444 **4.3 Sensitivity analysis**

445 In this section, sensitivity analysis is performed to show how changing the K factor, D factor, and
446 operating speed of ATMA vehicles can influence the PTSF during maintenance. As a result, the ATMA's
447 ODD is also changed. According to the HCM (HCM 2010), the range of the D factor is set as 0.5~0.65,
448 with the K factor as 0.09 to 0.12. The operating speeds of ATMA vehicles range from 5~15 mph. Thus,
449 The ODD of the ATMA can be analyzed in accordance with the results of sensitivity analysis.

Figure 9 presents the proposed relationship between traffic demand (i.e., AADT), and PTSF when factor D changes from 0.5 to 0.65. Figure 9 shows that the PTSF increases slightly as the D factor increases from 0.5 to 0.65. When the D factor is set as 0.5, the PTSF caused by a general truck platoon and an ATMA vehicle platoon increases from 23.6% to 69.5% as the AADT is increasing from 5,000 veh/day to 12,000 veh/day. The LOS decreases from A to D. As the D factor changes to 0.65, the PTSF increases from 26.2% to 72.5%, while the LOS decreases from A to D. We can conclude that, when the D factor increases from 0.5, to 0.55, 0.6, and then to 0.65, the corresponding AADT thresholds to define ATMA ODD are 11,500 veh/day, 11,500 veh/day, 11,000 veh/day, and 11,000 veh/day, respectively.

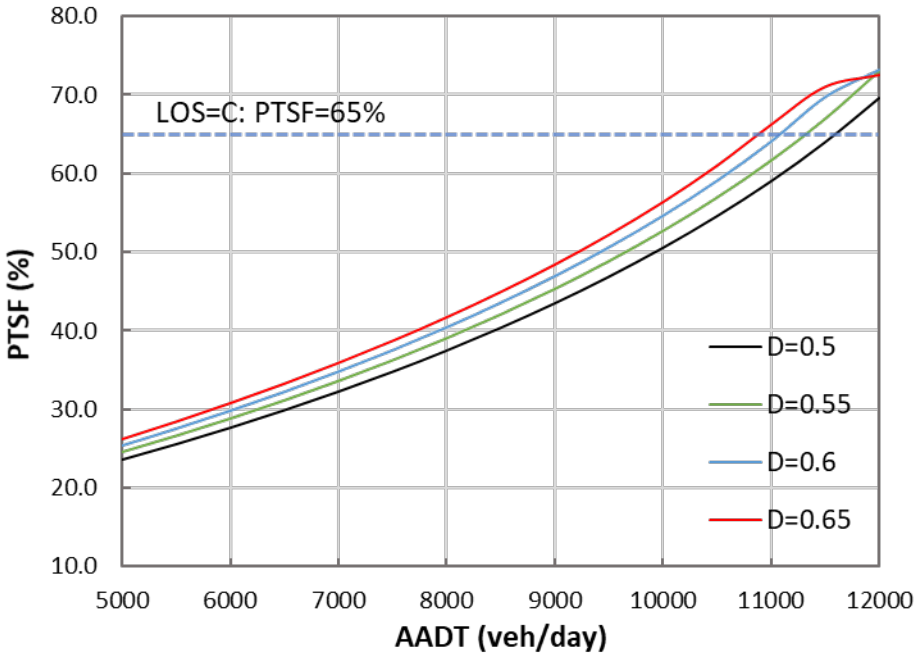


Figure 9 Proposed relationship between AADT and PTSF when D changes

Figure 10 presents the proposed relationship between traffic demand (i.e., AADT), and PTSF when factor K changes from 0.09 to 0.12. Figure 10 shows that the PTSF increases as the K factor increases from 0.09 to 0.12. When the K factor is set as 0.09, the PTSF caused by a general truck platoon and an ATMA vehicle platoon increases from 24.5% to 73.0% as the AADT is increasing from 5,000 veh/day to 12,000 veh/day. The LOS decreases from A to D. As K factor changes to 0.12, the PTSF increases from 31.9% to 91.5%, while the LOS decreases from A to e. We can conclude that, when K factor increases from 0.09,

to 0.10, 0.11, and then to 0.12, the corresponding AADT thresholds to define ATMA ODD are 11,500 veh/day, 10,500 veh/day, 9,500 veh/day, and 8,500 veh/day, respectively.

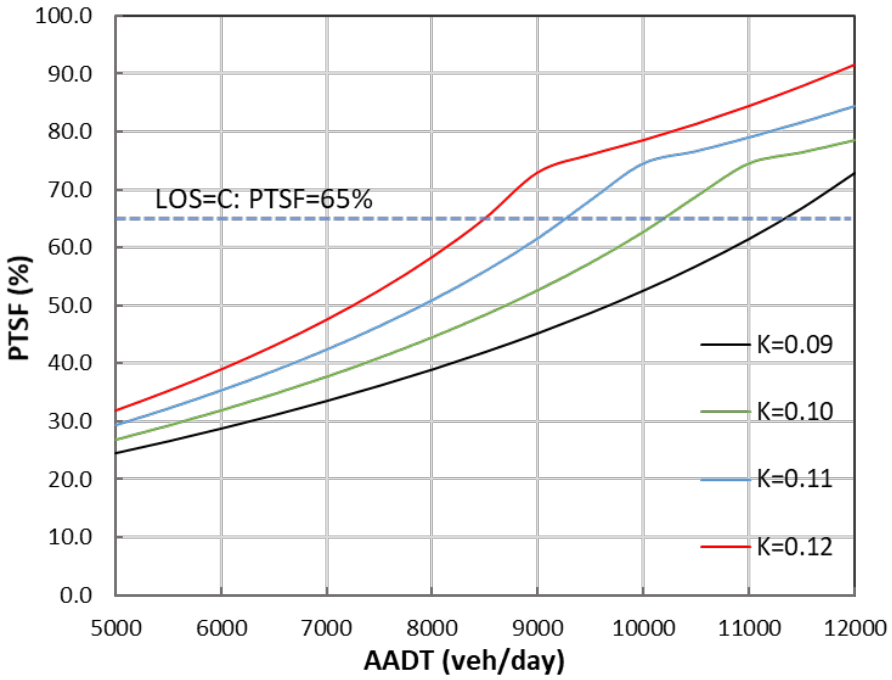


Figure 10 Proposed relationship between AADT and PTSF when K changes.

Figure 11 presents the proposed relationship between traffic demand (i.e., AADT), and PTSF when operating speeds of ATMA vehicles change from 5 mph to 15 mph. Figure 11 shows that the PTSF increases as the operating speed decreases from 15 mph to 5 mph. When the operating speed is 15 mph, the PTSF, caused by a general truck platoon and an ATMA vehicle platoon increases from 24.5% to 73.0%, as AADT increases from 5,000 veh/day to 12,000 veh/day. The LOS decreases from A to D. As the operating speed drops to 5 mph, the PTSF increases from 30.7% to 73.8%, while the LOS decreases from A to D. We can conclude that, when the operating speed decreases from 15 mph to 10 mph, and then to 5 mph, the corresponding AADT thresholds to define ATMA ODD are 11,000 veh/day, 10,000 veh/day, and 8,500 veh/day, respectively. The ODD turns out to be very sensitive to the operating speed of ATMA vehicles.

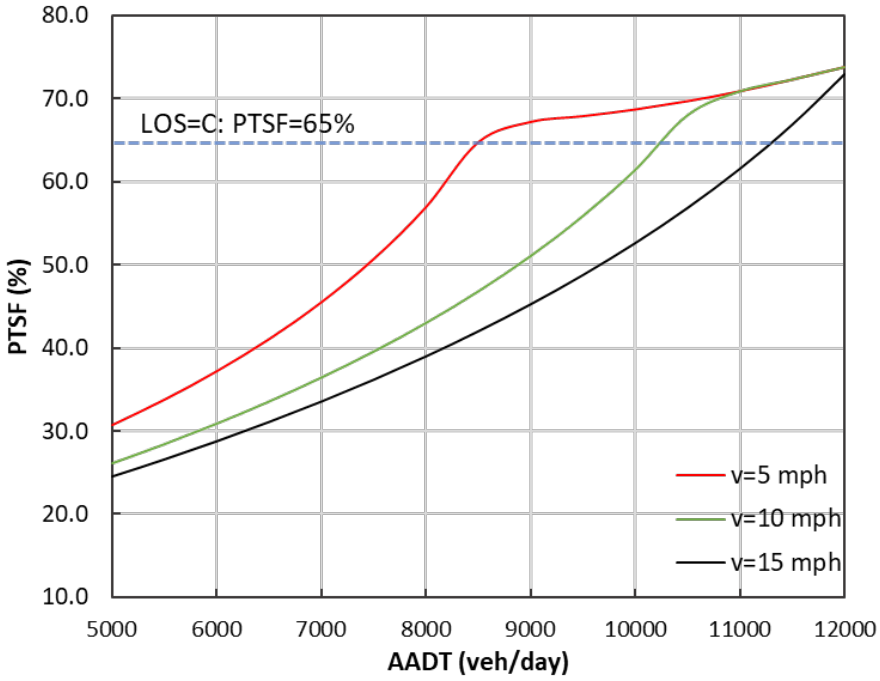


Figure 11 Proposed relationship between AADT and PTSF when the operating speed of an ATMA changes.

5. CONCLUSION

This manuscript aims to identify the traffic ODD of the ATMA vehicle system, which is applied to maintain transportation infrastructure at woke zones on two-lane highways. An analytical model was proposed to derive the PTSF affected by ATMA vehicles. As a key indicator of a two-lane highway's LOS, the PTSF becomes the linkage between demand (e.g., AADT) and LOS. Based on such a relationship, numerical analysis is performed to quantitively determine the traffic ODD of the ATMA system. The proposed model was validated by comparing with the HCM model. In this manuscript, we use LOS=C as a desirable design objective. Modeling result analysis shows that the ATMA vehicle system should perform maintenance work when the AADT is lower than 11,000 veh/day under a typical scenario. This AADT threshold turns to be sensitive to K factor, D factor, and ATMA operating speed. As the K (or D) factor increases or the operating speed decreases, the AADT threshold will drop accordingly.

6. ACKNOWLEDGEMENT

This material is based upon work supported by the Colorado DOT, and the AMT pool fund under the award number 5380-19-03, as well as USDOT through Mid-America Transportation Center, under the contract number 69A3551747107.

7. AUTHOR CONTRIBUTIONS

The authors confirm contribution to the paper, as follows: study conception and design: Q. Tang, X. Hu; methodology development: Q. Tang, X. Hu; analysis and interpretation of results: Q. Tang; draft manuscript preparation: Q. Tang, X. Hu. Manuscript review and editing: X. Hu. All authors reviewed the results and approved the final version of the manuscript.

8. REFERENCES

M. Cohen and A. Polus (2011). "Estimating percent-time-spent-following on two-lane rural highways." Transportation research part C: emerging technologies **19**(6): 1319-1325.

S. Descant. (2017). "Colorado DOT Launches Autonomous Vehicles to Improve Worker Safety." from https://www.govtech.com/fs/data/Colorado-DOT-Launches-Autonomous-Vehicles-to-Improve-Worker-Safety.html.

M. P. Dixon, S. S. K. Sarepali and K. A. Young (2002). "Field evaluation of highway capacity manual 2000 analysis procedures for two-lane highways." Transportation research record **1802**(1): 125-132.

FHWA (2010). Highway Capacity Manual, Transportation Reseach Board.

R. Harb, E. Radwan, X. Yan, A. Pande and M. Abdel-Aty (2008). "Freeway Work-Zone Crash Analysis and Risk Identification Using Multiple and Conditional Logistic Regression." Journal of Transportation Engineering **134**(5): 203-214.

HCM (2010). "Highway Capacity Manual." Transportation Research Board, Washington, DC: 1207.

J. A. Laval (2006). "A macroscopic theory of two-lane rural roads." Transportation Research Part B: Methodological **40**(10): 937-944.

R. T. Luttinen (2001). "Percent time-spent-following as performance measure for two-lane highways." Transportation Research Record **1776**(1): 52-59.

Q. Ma, H. Yang, K. Xie, Z. Wang and X. Hu (2019). "Taxicab crashes modeling with informative spatial autocorrelation." Accident Analysis & Prevention **131**: 297-307.

Q. Meng and J. Weng (2011). "Evaluation of rear-end crash risk at work zone using work zone traffic data." Accident Analysis & Prevention **43**(4): 1291-1300.

MoDOT (2018). Leader-Follower Truck Mounted Attenuator System Request for Proposal. Jefferson City, Missouri, Missouri Department of Transportation.

G. F. Newell (1998). "A moving bottleneck." Transportation Research Part B: Methodological **32**(8): 531-537.

NHTSA (2016). Federal automated vehicles policy: Accelerating the next revolution in roadway safety, US Department of Transportation.

A. Polus and M. Cohen (2009). "Theoretical and empirical relationships for the quality of flow and for a new level of service on two-lane highways." Journal of Transportation Engineering **135**(6): 380-385.

M. Pursula (1995). "Approximation of percentage time delay with local measurements." Transportation research record: 58-65.

Q. Tang, Y. Cheng, X. Hu, C. Chen, Y. Song and R. Qin (2021). "Evaluation Methodology of Leader-Follower Autonomous Vehicle System for Work Zone Maintenance." Transportation Research Record **0**(0): 0361198120985233.

Q. Tang, X. Hu and R. Qin (2021). "Development of Operation Guidelines for Leader-Follower Autonomous Maintenance Vehicles at Work Zone Locations." Transportation Research Record Journal of the Transportation Research Board.